

Classes with Attributes and Methods

1. User (Abstract Base Class)

Purpose: Base class for all user types in the system.

Attributes

- **userId:** Long
- **email:** String
- **passwordHash:** String
- **fullName:** String
- **role:** RoleEnum
- **createdAt:** DateTime
- **isVerified:** Boolean
- **phoneNumber:** String

Methods

- **+ login():** Boolean
- **+ logout():** void
- **+ resetPassword(email: String):** Boolean
- **+ updateProfile(data: UserData):** Boolean
- **+ verifyEmail(token: String):** Boolean
- **# hashPassword(password: String):** String
- **- validateEmail(email: String):** Boolean

Visibility Key:

- **+ Public:** Accessible from anywhere.
 - **# Protected:** Accessible by subclasses only.
 - **- Private:** Only accessible within this class.
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2. Customer (Extends User)

Purpose: Represents end-users who create tickets.

Attributes

- **customerId:** Long
- **satisfactionRating:** Float
- **totalTickets:** Integer
- **preferredLanguage:** String

Methods

- **+ createTicket(title: String, description: String):** Ticket
 - **+ viewMyTickets():** List<Ticket>
 - **+ chatWithSupport(ticketId: Long, message: String):** Message
 - **+ rateSupport(ticketId: Long, rating: Integer):** void
 - **+ searchTickets(query: String):** List<Ticket>
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3. SupportAgent (Extends User)

Purpose: Staff members who handle customer tickets.

Attributes

- **agentId:** Long
- **department:** String
- **avgResponseTime:** Float
- **ticketsResolved:** Integer
- **performanceRating:** Float
- **availability:** Boolean

Methods

- **+ viewTicketQueue():** List<Ticket>
- **+ acceptTicket(ticketId: Long):** Boolean

- + **respondToTicket(ticketId: Long, response: String):** Message
 - + **updateTicketStatus(ticketId: Long, status: StatusEnum):** Boolean
 - + **reassignTicket(ticketId: Long, agentId: Long):** Boolean
 - + **addInternalNote(ticketId: Long, note: String):** void
 - + **viewPerformance():** AgentMetrics
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4. Administrator (Extends User)

Purpose: System managers with full access.

Attributes

- **adminId:** Long
- **accessLevel:** Integer
- **lastLoginTime:** DateTime

Methods

- + **createUser(userData: UserData):** User
 - + **assignRole(userId: Long, role: RoleEnum):** Boolean
 - + **uploadDocument(file: File):** Document
 - + **deleteUser(userId: Long):** Boolean
 - + **viewAnalytics():** Dashboard
 - + **generateReport(type: String, dateRange: DateRange):** Report
 - + **configureSettings(settings: SystemConfig):** Boolean
 - + **viewAuditLogs():** List<AuditLog>
 - + **manageKnowledgeBase():** void
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5. Ticket

Purpose: Represents a customer support request.

Attributes

- **ticketId:** Long
- **title:** String
- **description:** String
- **category:** String
- **priority:** PriorityEnum
- **status:** StatusEnum
- **createdAt:** DateTime
- **updatedAt:** DateTime
- **resolvedAt:** DateTime
- **customerId:** Long
- **assignedAgentId:** Long

Methods

- + **create():** Boolean
 - + **updateStatus(status: StatusEnum):** Boolean
 - + **assignToAgent(agentId: Long):** Boolean
 - + **close():** Boolean
 - + **reopen():** Boolean
 - + **getHistory():** List<TicketHistory>
 - + **addTag(tag: String):** void
 - - **autoClose():** void
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6. Message

Purpose: Chat messages between customer, agent, and AI.

Attributes

- **messageId:** Long
- **ticketId:** Long
- **senderId:** Long
- **senderType:** SenderTypeEnum
- **messageContent:** String
- **isAiGenerated:** Boolean
- **confidenceScore:** Float
- **timestamp:** DateTime
- **isRead:** Boolean

Methods

- + **send():** Boolean
 - + **markAsRead():** void
 - + **getTimestamp():** DateTime
 - + **isFromAI():** Boolean
 - - **validateContent():** Boolean
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7. AIBot

Purpose: AI chatbot that generates automated responses.

Attributes

- **botId:** Long
- **modelName:** String
- **apiKey:** String
- **confidenceThreshold:** Float

- **responseTime:** Float

Methods

- + **generateResponse(query: String):** AIResponse
 - + **categorizeTicket(description: String):** String
 - + **calculateConfidence(response: String):** Float
 - + **searchKnowledgeBase(query: String):** List<Document>
 - - **embedQuery(query: String):** Vector
 - - **rankDocuments(documents: List<Document>):** List<Document>
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8. KnowledgeBase

Purpose: Repository of company documents for AI.

Attributes

- **kbId:** Long
- **name:** String
- **totalDocuments:** Integer
- **lastUpdated:** DateTime

Methods

- + **addDocument(document: Document):** Boolean
 - + **removeDocument(docId: Long):** Boolean
 - + **searchDocuments(query: String):** List<Document>
 - + **indexDocument(document: Document):** Boolean
 - + **getDocument(docId: Long):** Document
 - - **createEmbeddings(text: String):** Vector
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9. Document

Purpose: Individual document in the knowledge base.

Attributes

- **docId:** Long
- **title:** String
- **filePath:** String
- **fileType:** String
- **uploadedBy:** Long
- **uploadedAt:** DateTime
- **isIndexed:** Boolean
- **content:** String

Methods

- **+ upload():** Boolean
 - **+ delete():** Boolean
 - **+ extract():** String
 - **+ split():** List<String>
 - **- validate():** Boolean
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10. Notification

Purpose: Email/SMS alerts to users.

Attributes

- **notificationId:** Long
- **recipientId:** Long
- **type:** NotificationTypeEnum
- **subject:** String
- **body:** String

- **sentAt:** DateTime
- **deliveryStatus:** String

Methods

- + **send():** Boolean
 - + **retry():** Boolean
 - + **getStatus():** String
 - - **formatMessage():** String
 - - **validateRecipient():** Boolean
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11. Report

Purpose: Analytics and system reports.

Attributes

- **reportId:** Long
- **reportType:** String
- **generatedBy:** Long
- **generatedAt:** DateTime
- **dateRange:** DateRange
- **data:** JSON

Methods

- + **generate():** Boolean
- + **exportToPDF():** File
- + **exportToCSV():** File
- + **getMetrics():** Map<String, Object>
- - **calculateStatistics():** void